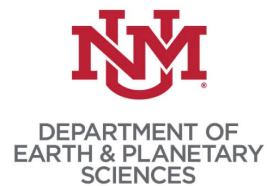


Pebble accretion: Earth's composition and water origin

Susmita Garai, Zachary D. Sharp, Peter L. Olson

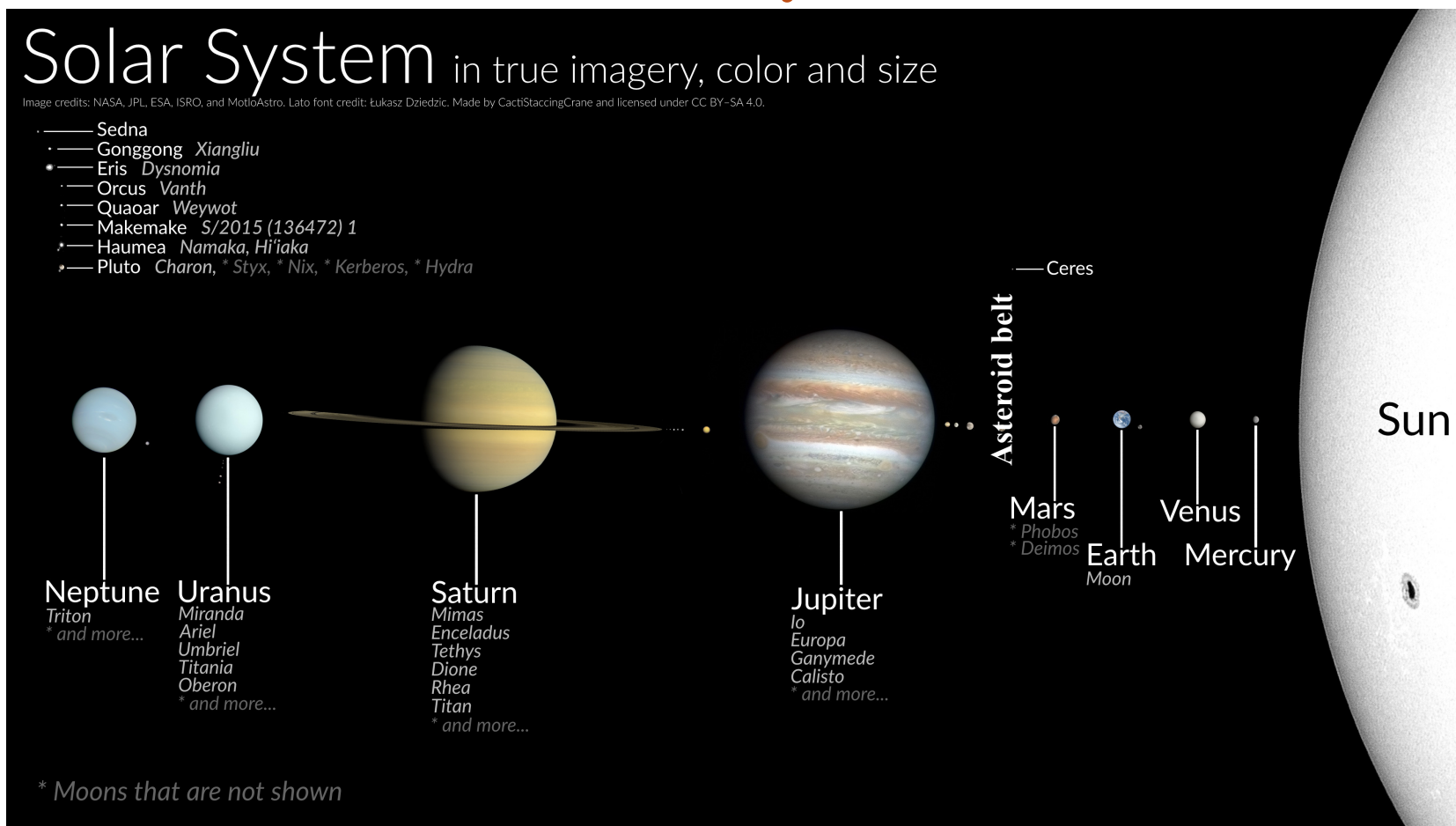
Earth and Planetary Sciences, University of New Mexico



sgarai@unm.edu

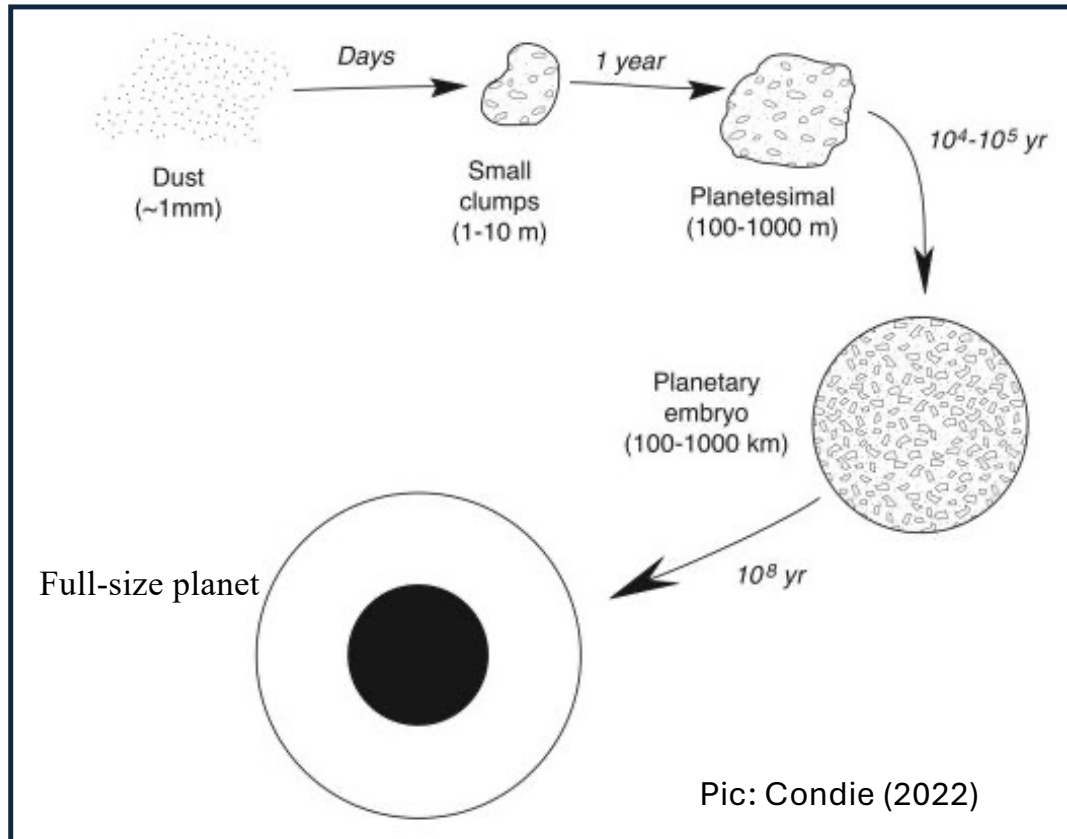


Our Solar system



How do terrestrial planets form?

Impact-based model



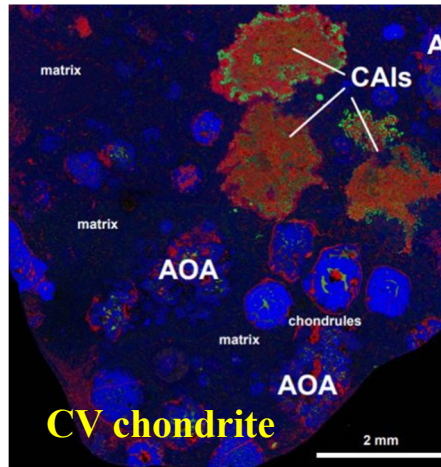
Growth of protoplanets by **stochastic collisions** between **km-sized** bodies (e.g., chondrites) over **10s-100s of millions of years**. (e.g., Safronov, 1969).

Earth's bulk composition with chondrites

Chondrites



(Grossman and Zipfel, 2001)



(Jean et al., 2019)

Earth's bulk composition with chondrites

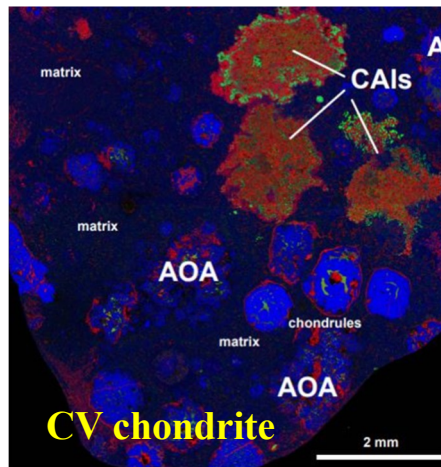
Chondrites

Chondrite model



O chondrite

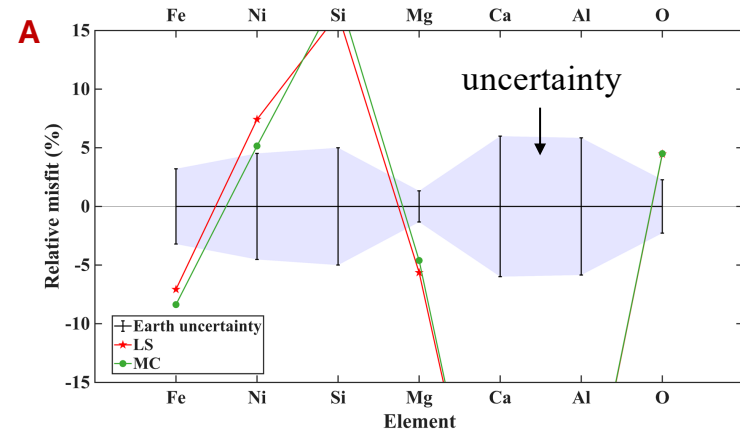
(Grossman and Zipfel, 2001)



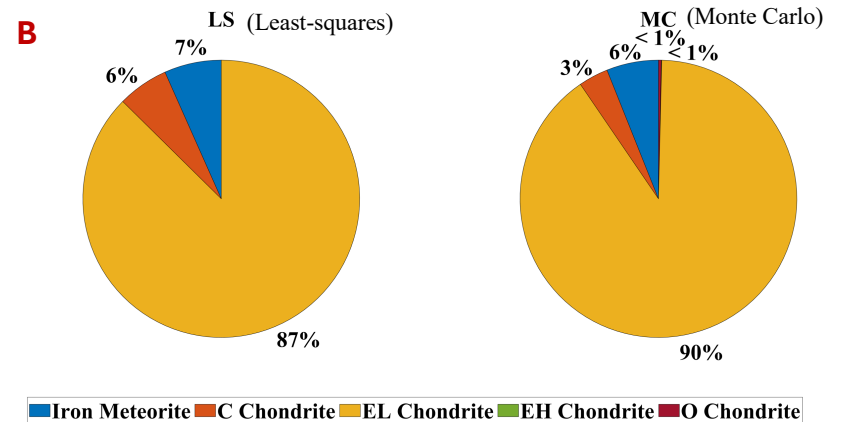
CV chondrite

(Jean et al., 2019)

Misfit=(Model-Earth)/Earth



Chondrite proportion



Earth's bulk composition with chondrites

Determining the possible building blocks of the Earth and Mars

Thomas H. BURBINE^{1*} and Kevin M. O'BRIEN²

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Abstract—To determine the possible building blocks of the Earth and Mars, 225,792,840 possible combinations of the bulk oxygen isotopic and chemical compositions of 13 chondritic groups at 5% mass increments were examined. Only a very small percentage of the combinations match the oxygen isotopic composition, the assumed bulk FeO concentration, and the assumed Fe/Al weight ratio for the Earth. Since chondrites are enriched in silicon relative to estimates of the bulk Earth, none of the combinations fall near the terrestrial magmatic fractionation trend line in Mg/Si-Al/Si space. More combinations match the oxygen isotopic composition and the assumed bulk FeO concentration for Mars. These combinations fall near the trend for shergottite meteorites in Mg/Si-Al/Si space. One explanation for the difficulty in forming Earth out of known chondrites is that the Earth may be composed predominately of material that did not survive to the present day as meteorites. Another explanation could be that significant amounts of silicon are sequestered in the core and/or lower mantle of the Earth.

Burbine and O'brien (2004)

The mixed chemical and isotopic affinities of the Earth's composition is a conundrum that has yet to be resolved. It also means that no model for the Earth's composition that uses known chondrite groups as building blocks can satisfy both the isotopic and elemental constraints. This is illustrated in Fig. 1 where the elemental composition of the best fit to the Earth's isotopic compositions determined by Dauphas (2017) (a mix of EC, OC and CO/CV materials) is very different from estimated by Alexander (2022) composition. Also based primarily on isotopes, Schner

teroids that are associated with noncarbonaceous chondrite meteorites (NC), mostly ordinary chondrites (OC) from the inner solar system (de Micas et al., 2022; Burkhardt et al., 2021; Kleine et al., 2020). [See Section 2 for more details.]

However, no chondrite has a composition equal to that of the Earth or other terrestrial planets, and none can be simply identified as approaching the composition of the Earth without ad hoc modifications to its composition to make the two match (McDonough (2025) Dauphas, 2017). The meteoritic r

Pebble accretion: the emerging paradigm

- ***Pebbles:*** aerodynamically small solids: sub-mm to cm-sized, meaning they are subject to gas drag (Weidenschilling, 1977). Pebbles drift toward the central star due to headwind from the gas.

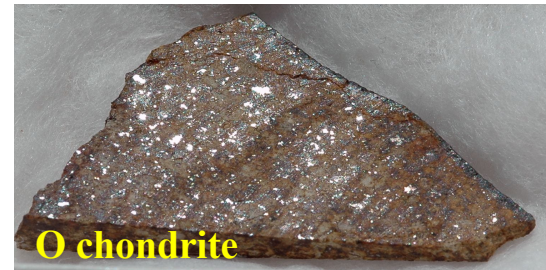
Pebble accretion: the emerging paradigm

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- ***Pebble accretion:*** A seed mass of ~ 100 km or more grows rapidly by gravitationally capturing pebbles that are drifting toward the central star. (e.g., Lambrechts
- & Johansen, 2012).

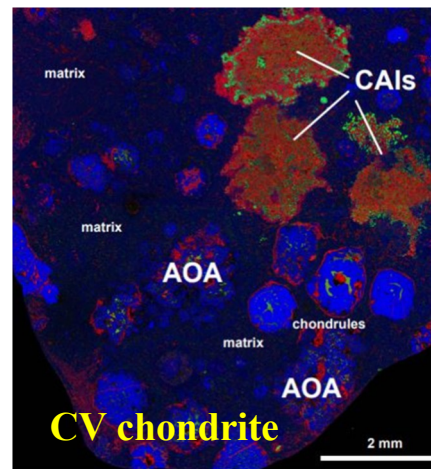
Pebble accretion: the emerging paradigm

Chondritic components

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(Grossman and Zipfel, 2001)



(Jean et al., 2019)

Components:
Chondrules, metal
grains, refractory
inclusions.

**They predate their
host chondrites.**

Do chondritic component pebbles

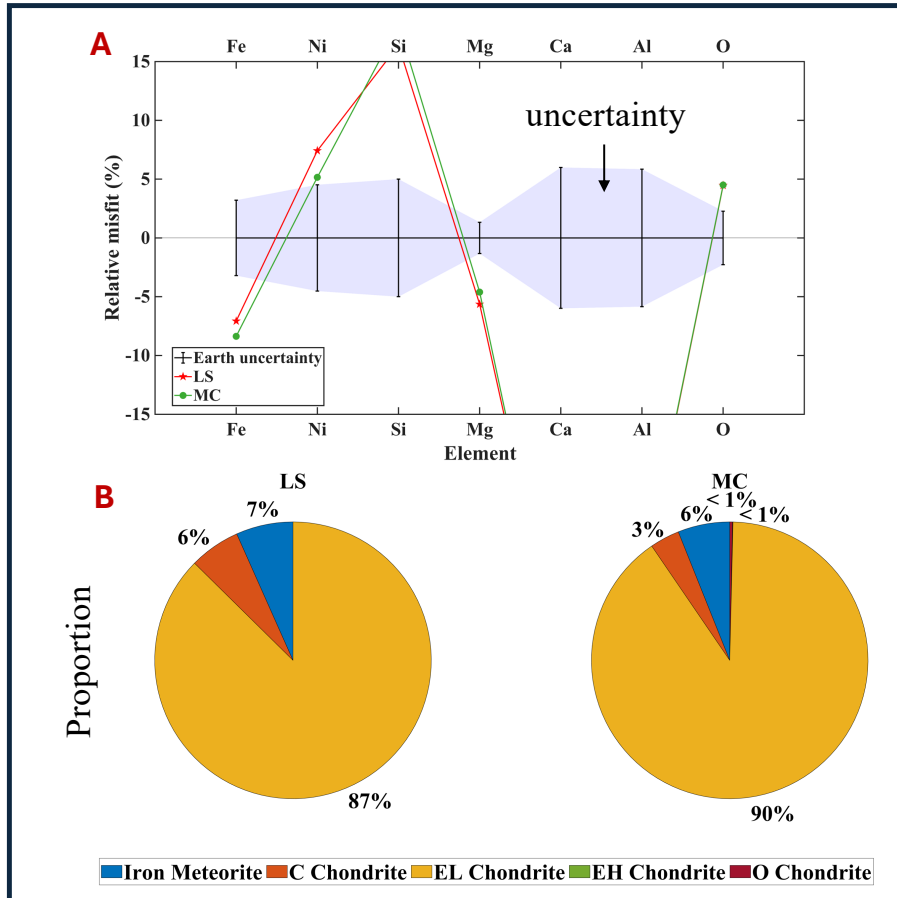
- 1. Match Earth's major element composition (Fe, Ni, Si, Mg, Ca, Al, O)?**
- 2. Deliver Earth's water?**
- 3. Match Earth's siderophile composition (Ni, Co, V, Cr, Mo, Mn, and W)?**
- 4. Match Earth's moderately volatile depletion (Si, Na, K, and Zn)?**

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Earth's major element composition with pebbles

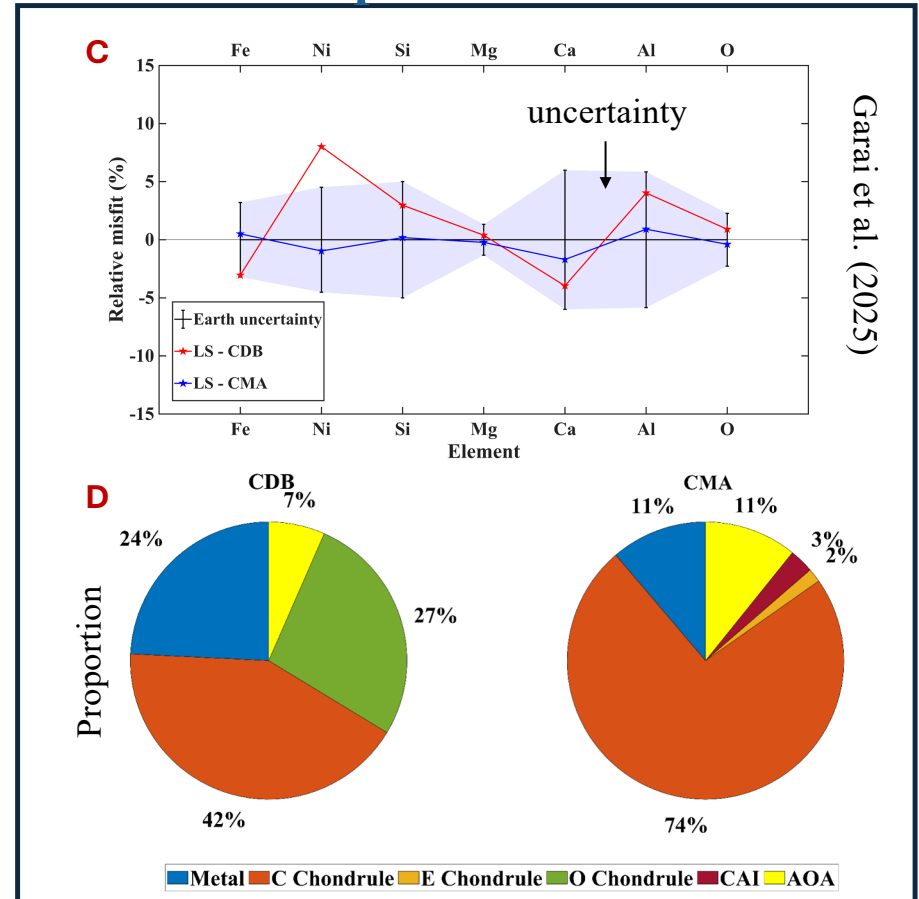
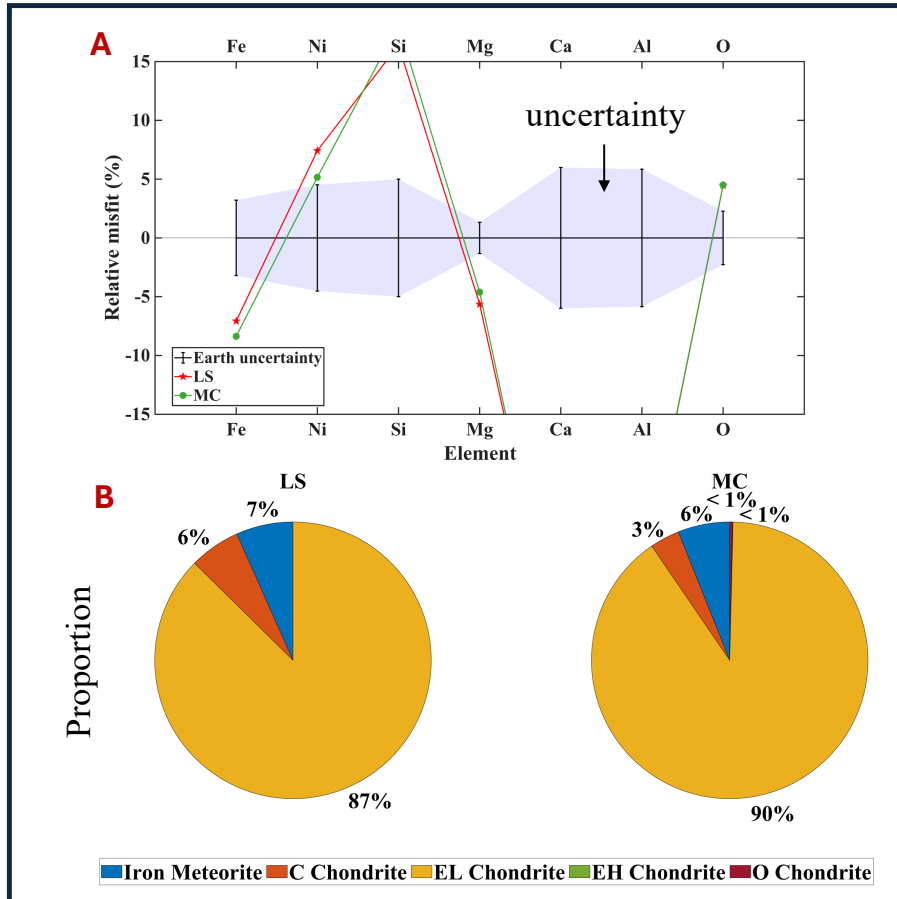
Chondrite model



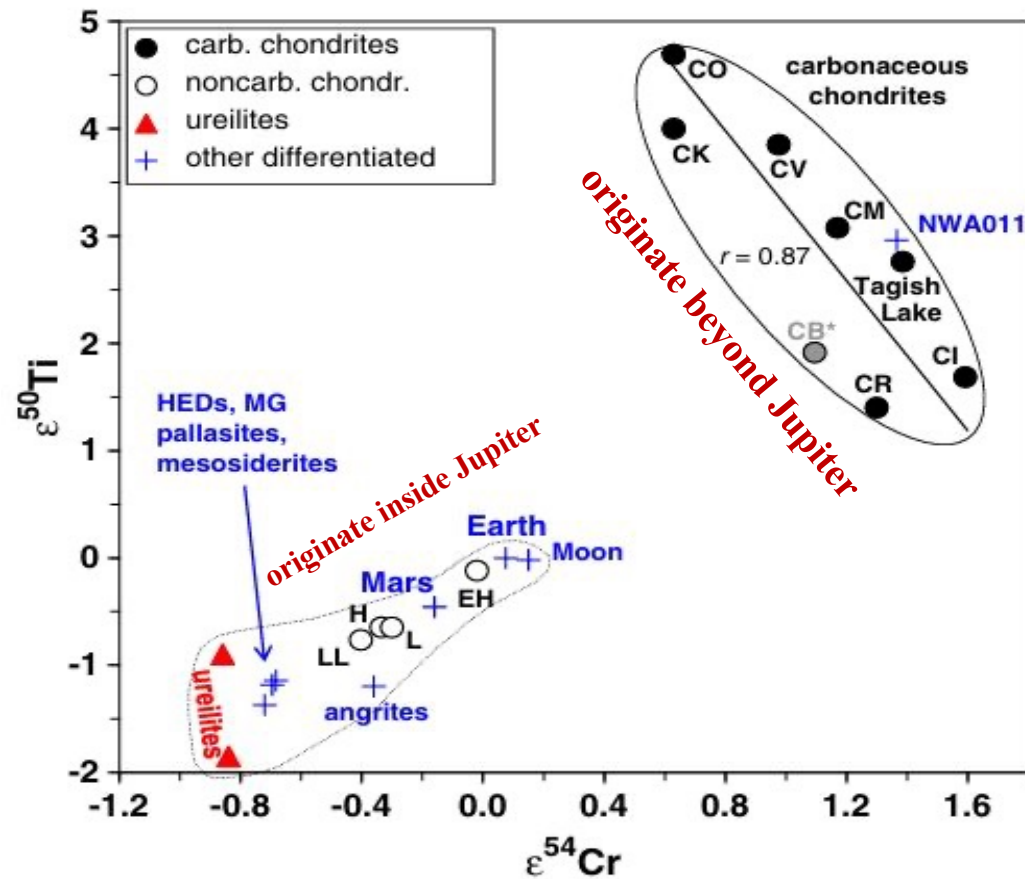
Earth's major element composition with pebbles

Chondrite model

Component model



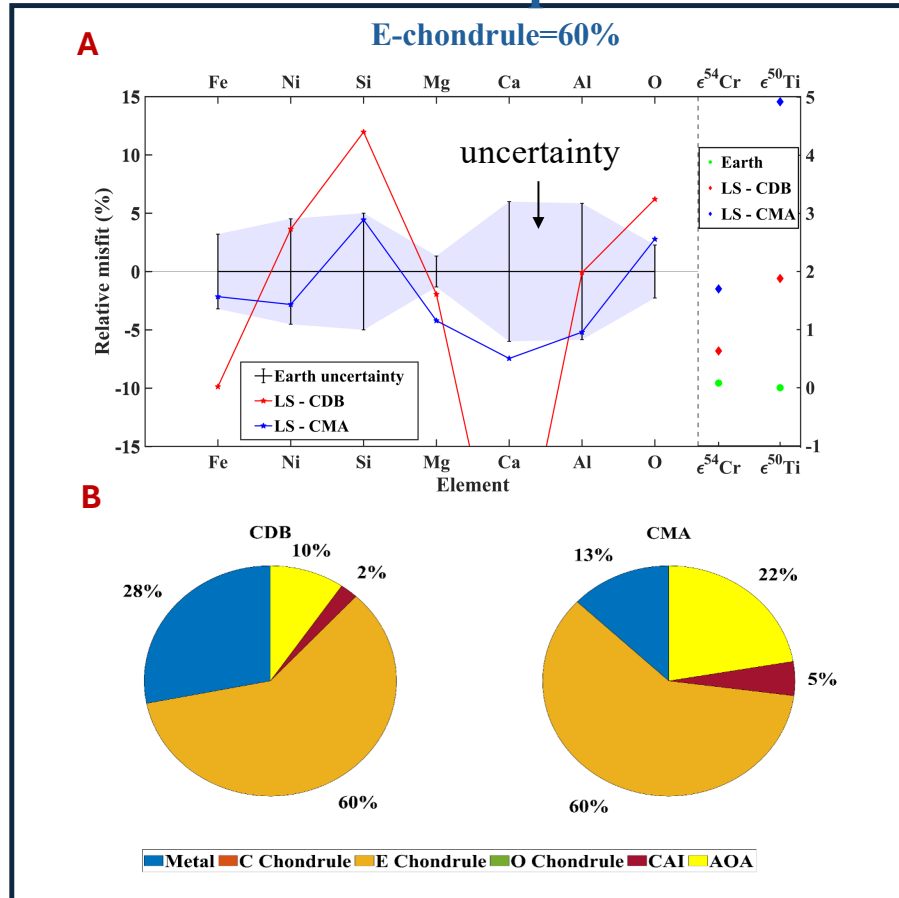
Earth's isotopic compositions require a large NC source



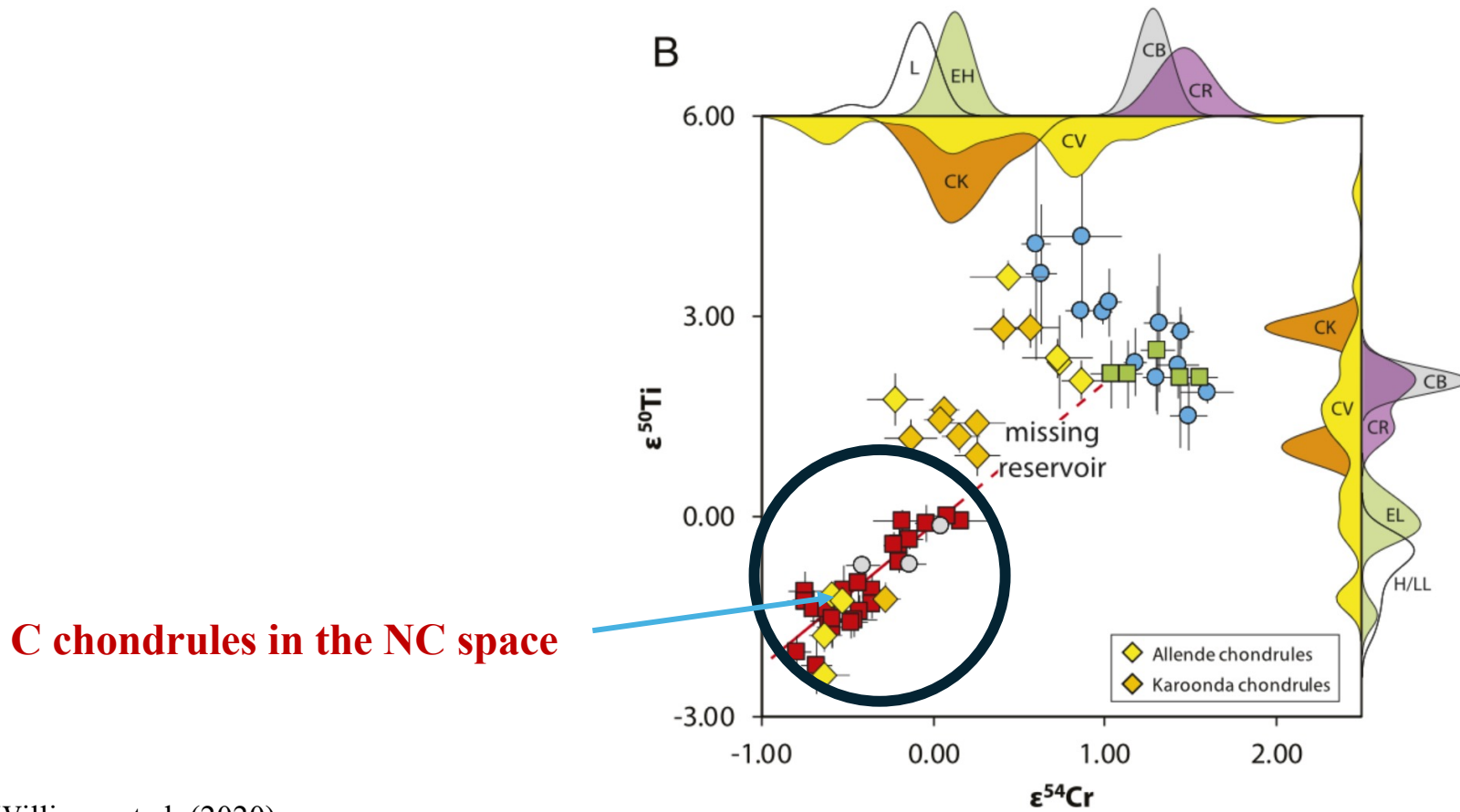
Warren (2011)

Earth's bulk composition with 60% E chondrules

Constrained Component model



C-chondrules found to have NC-like $\epsilon^{54}\text{Cr} / \epsilon^{50}\text{Ti}$

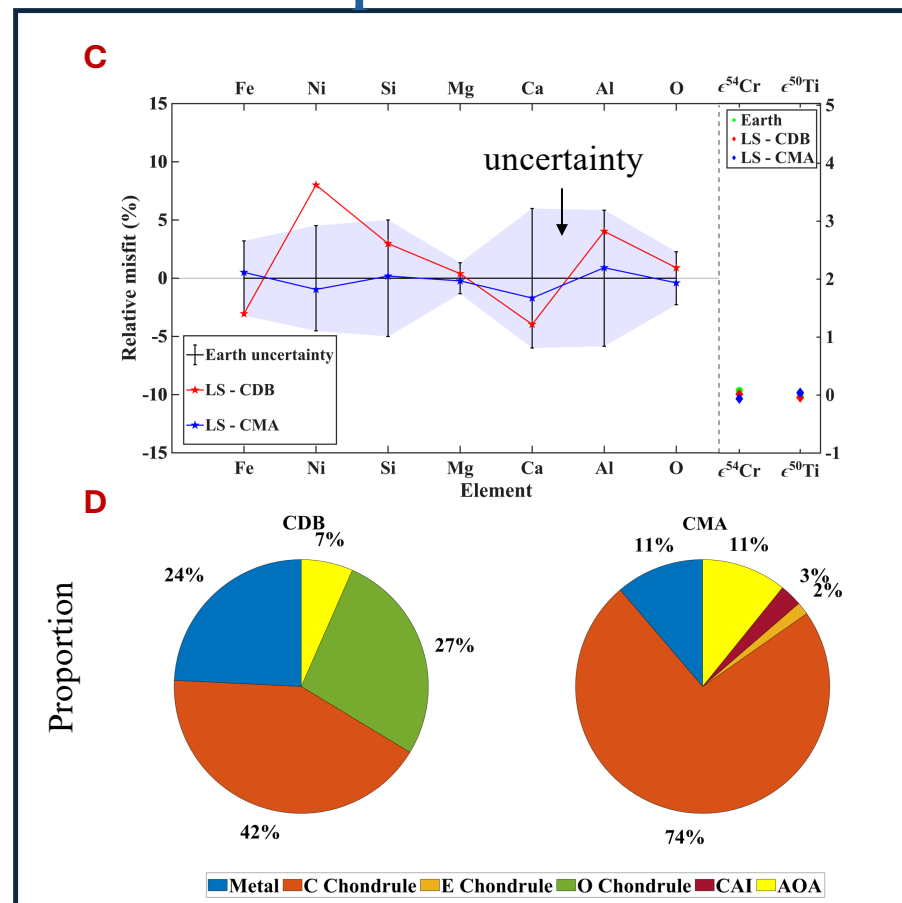
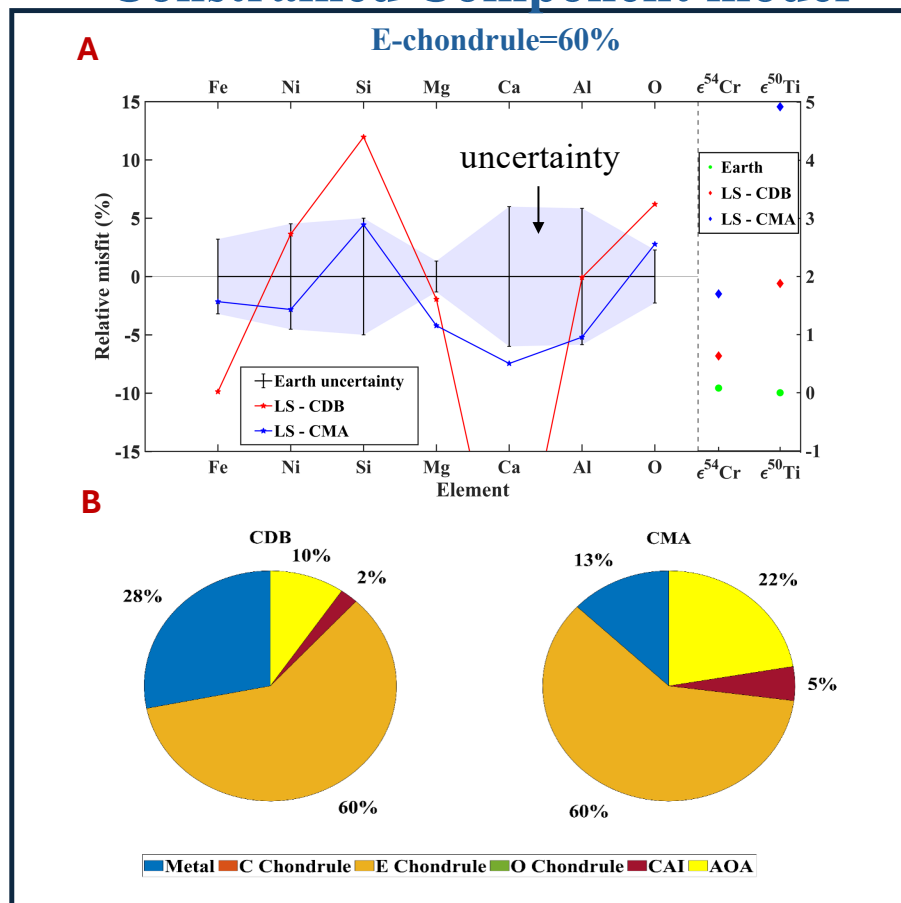


Williams et al. (2020)

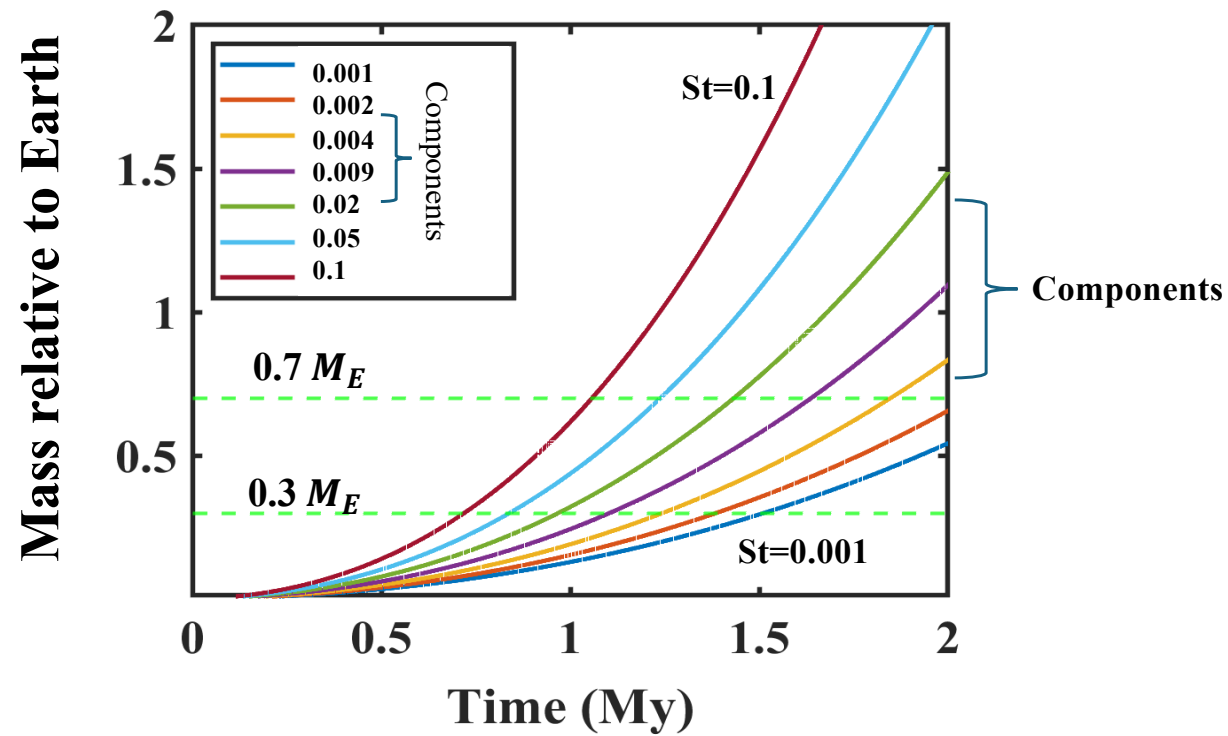
Earth's major +isotopic compositions with pebbles

Constrained Component model

Component model



Accretion timeline for Earth



Do chondritic component pebbles

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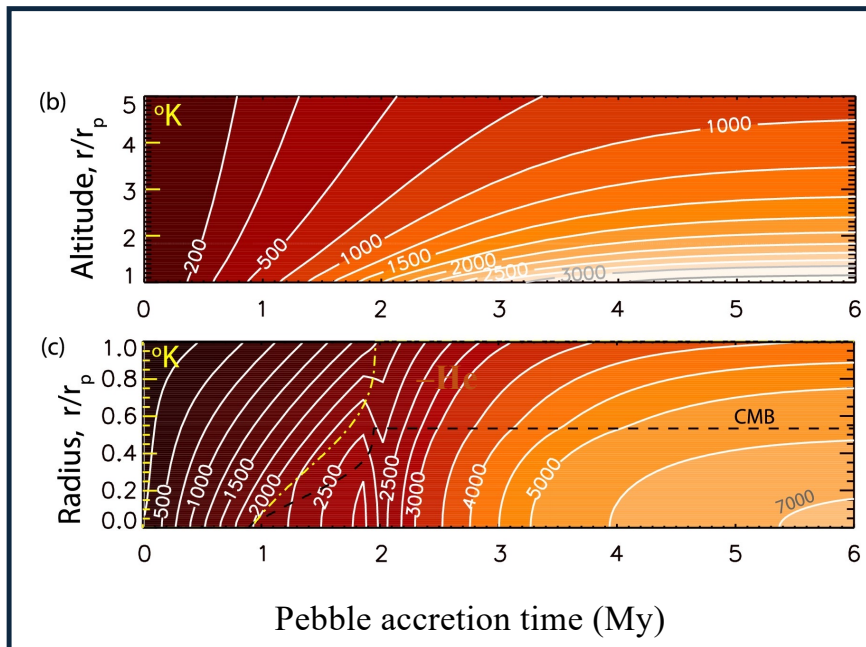
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Rapid accretion leads to a hot atmosphere

Earth's temperature

A

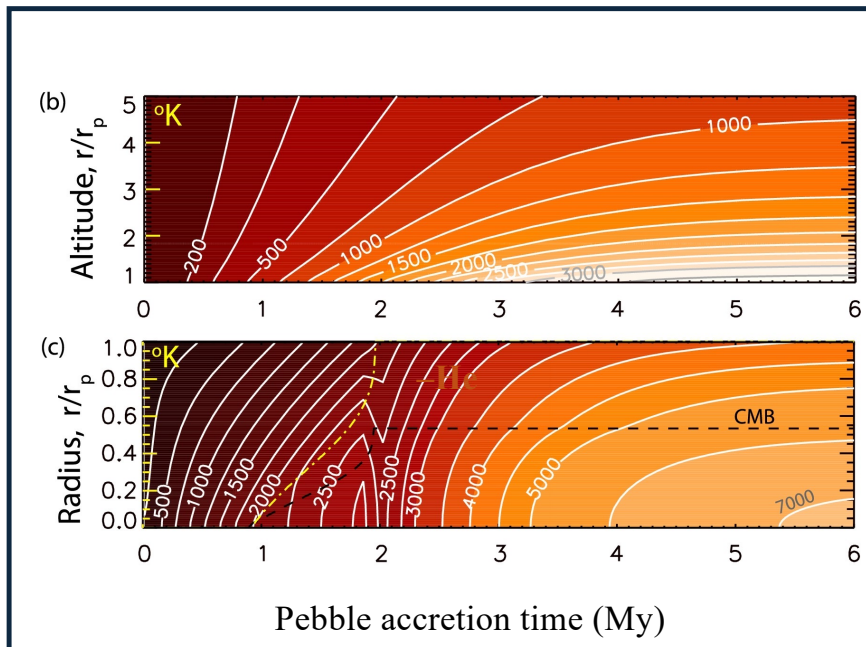


Olson et al. (2022)

Rapid accretion leads to a hot, H₂–He-rich atmosphere

Earth's temperature

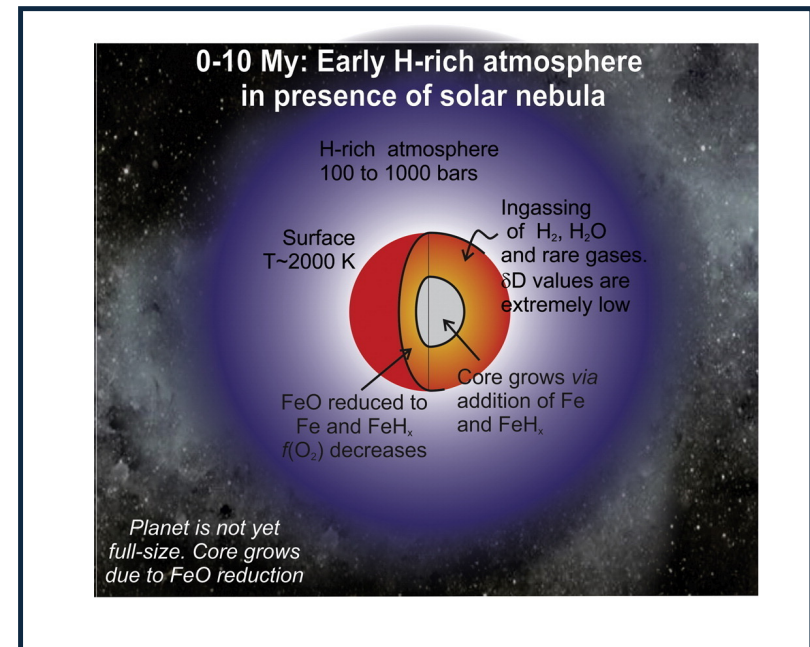
A



Olson et al. (2022)

Nebular ingassing

B



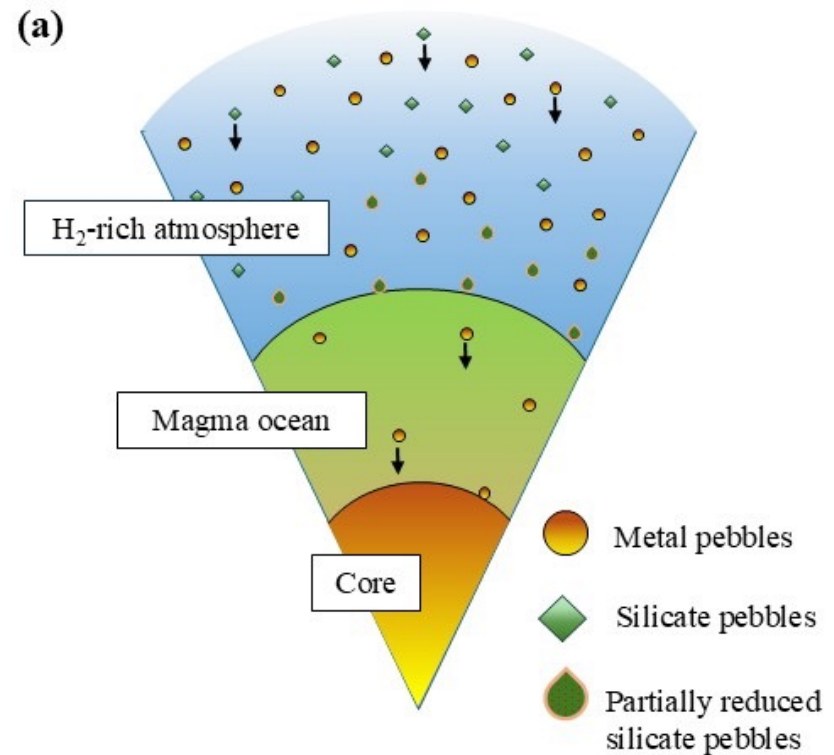
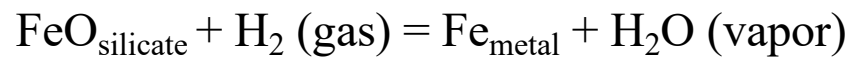
Sharp (2017)

Rapid accretion leads to a hot, H₂–He-rich atmosphere

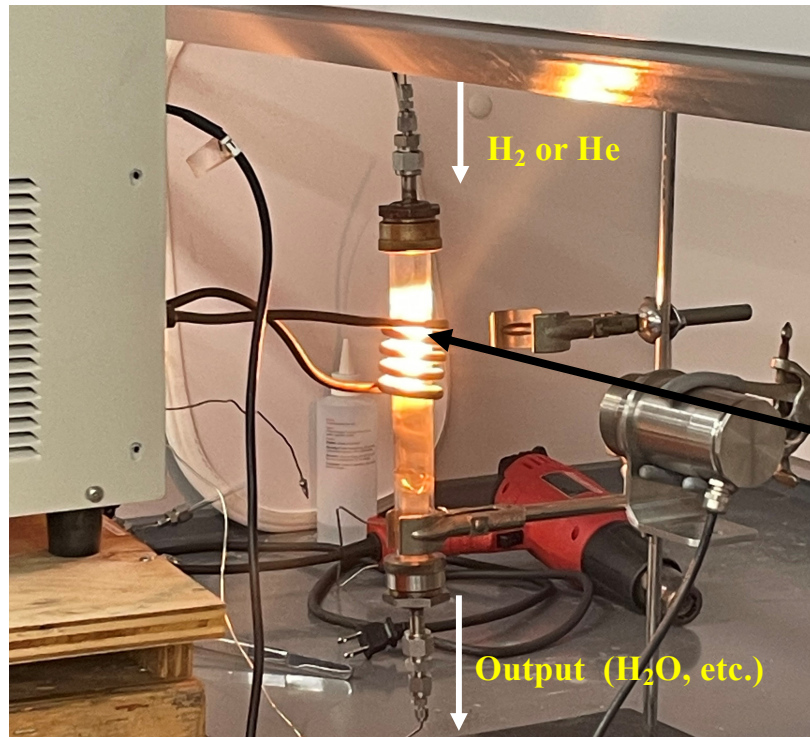
1. Pebble rain

2. Volatile loss from the settling pebbles

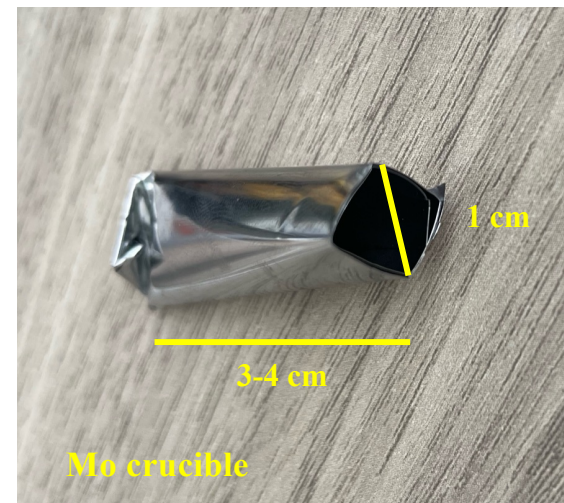
3. Water formation according to



Experimental simulation of pebble rain



Melted ultramafic compositions (e.g., MORB) at 1200-1400°C for 15 minutes to 3 hours in He and H₂ streams.

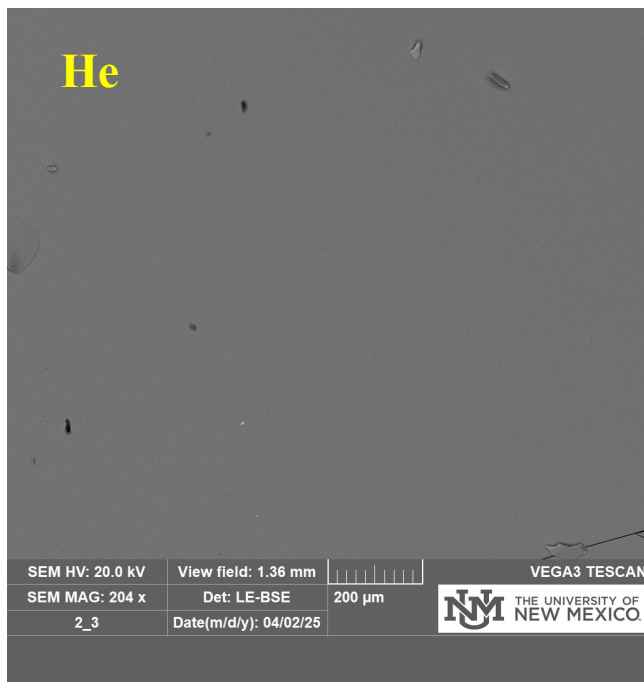


Garai et al. in review

Results: Backscattered electron (BSE) images

Light (Dark) ~ High (low) atomic number

A

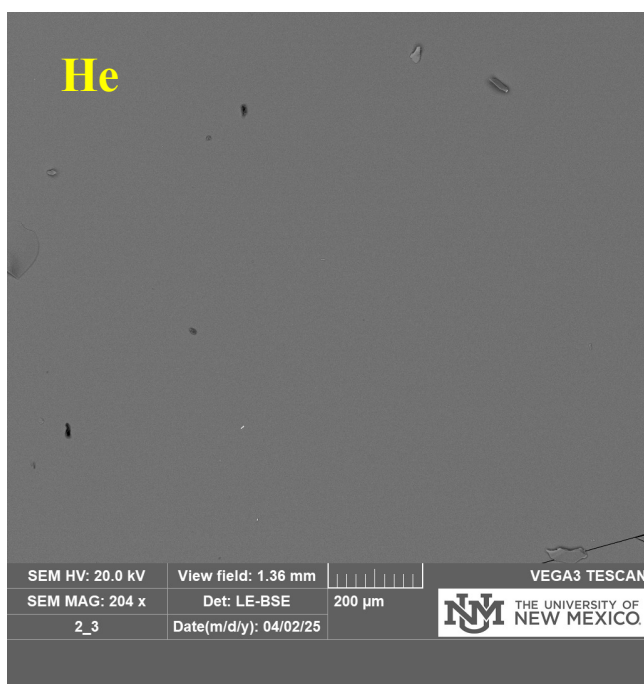


1300 °C, 3 hours

Results: Backscattered electron (BSE) images

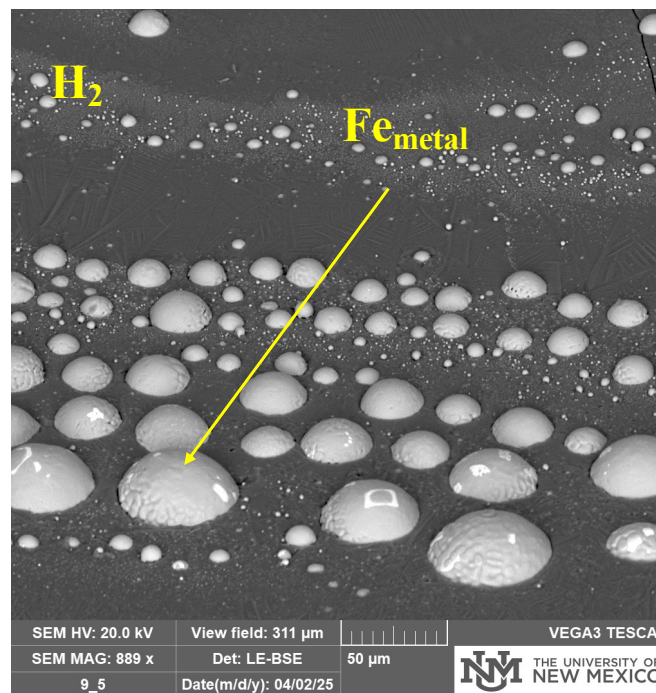
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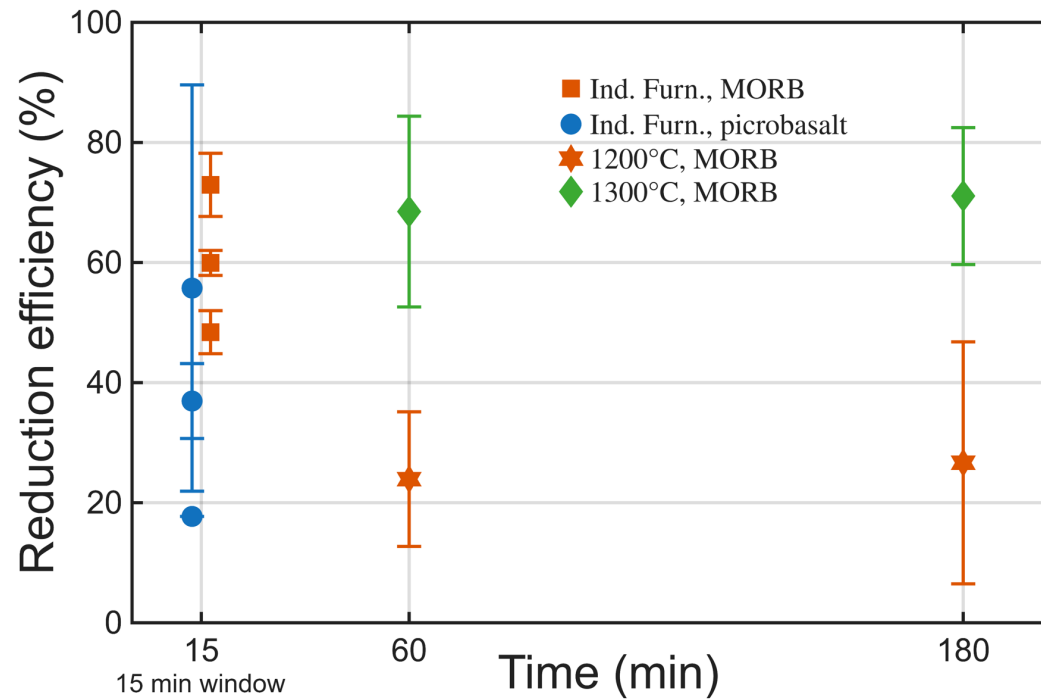


1300 °C, 3 hours

B



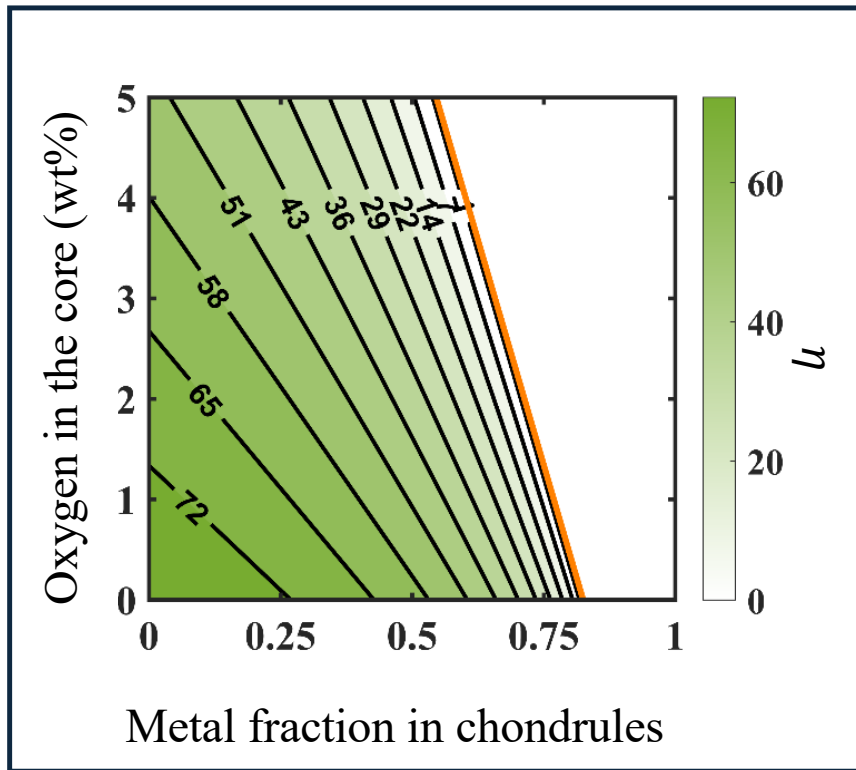
FeO reduction efficiency



The amount of FeO reduced to Fe metal after the reduction experiments.

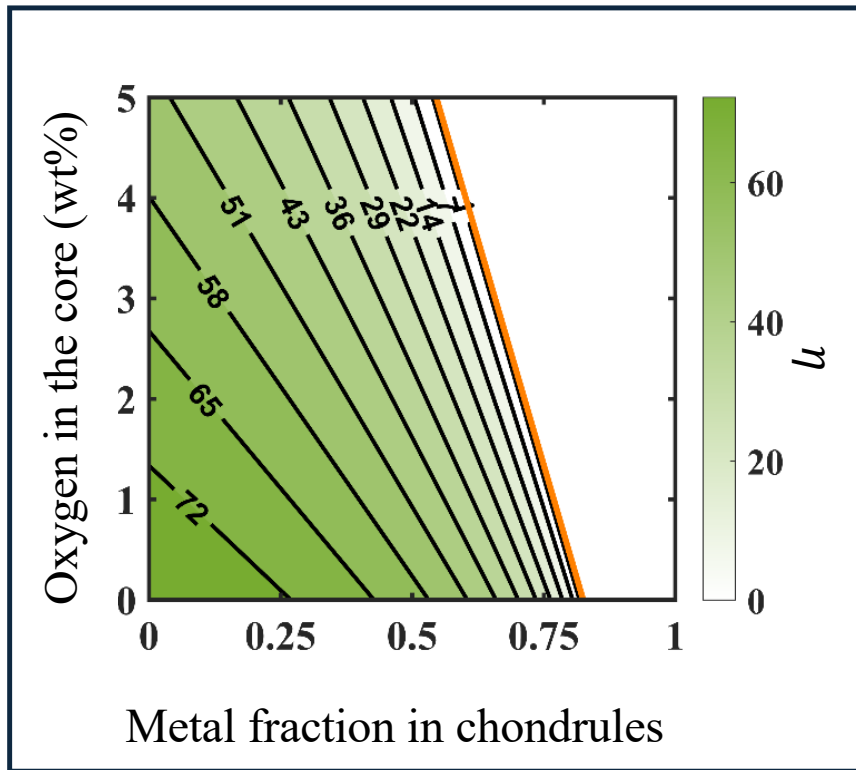
Water produced on Earth by FeO reduction

A FeO reduction efficiency (CMA)

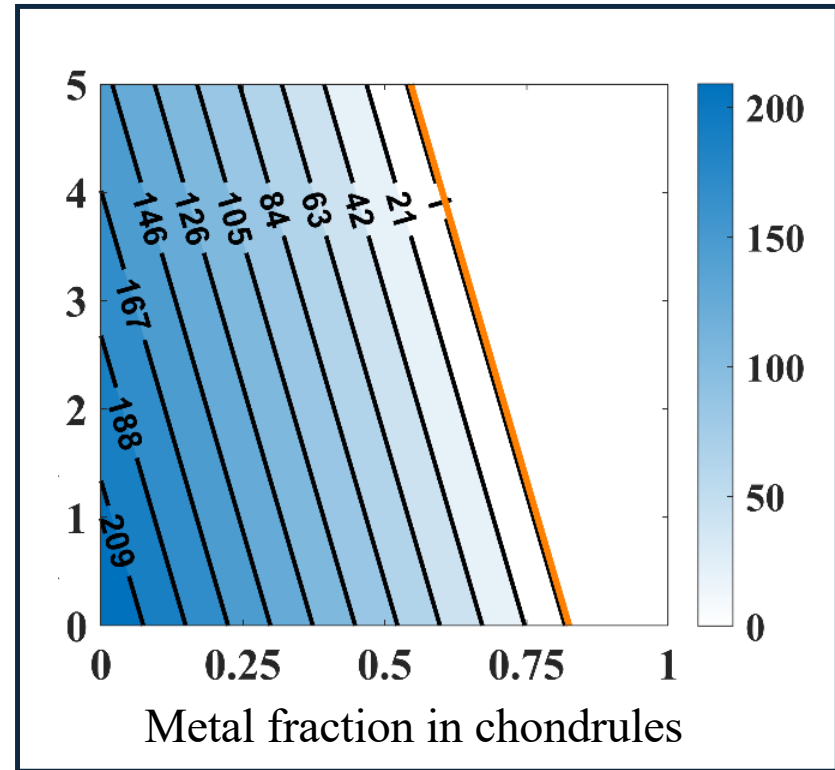


Water produced on Earth by FeO reduction

A FeO reduction efficiency (CMA)



B Water produced



Do chondritic component pebbles

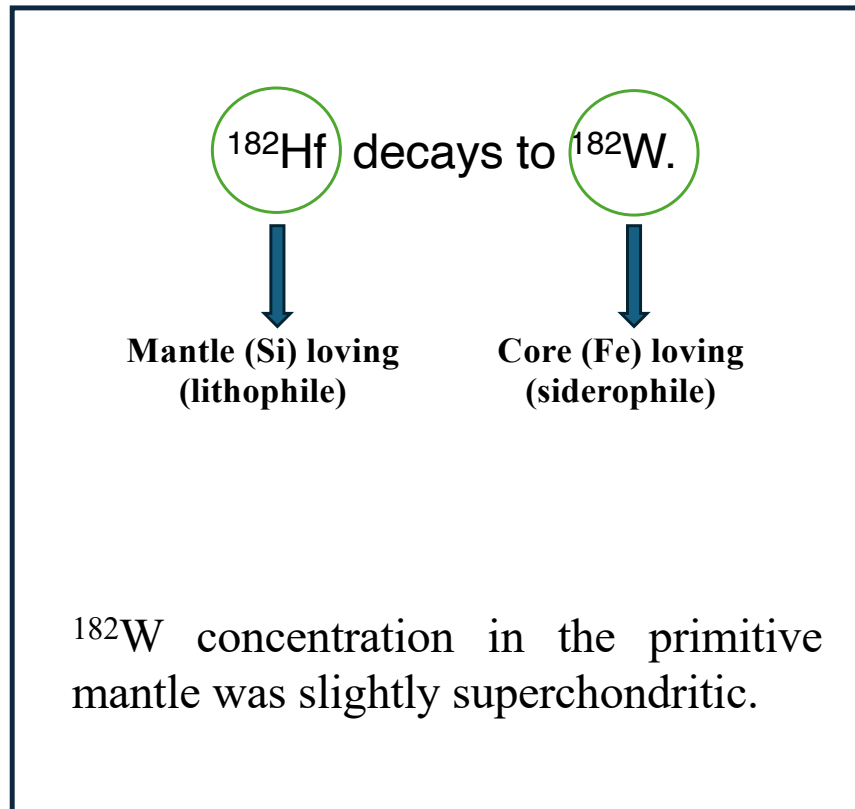
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4. Match Earth's moderately volatile depletion (Si (as SiO), Na, K, and Zn)? ☐

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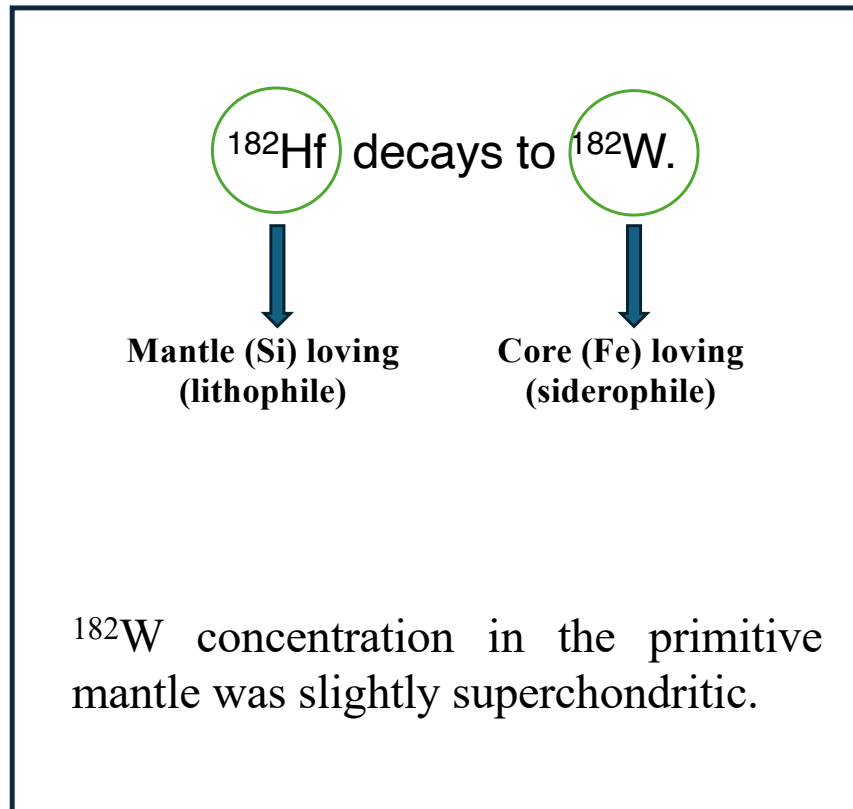
Earth's hafnium-tungsten system

Fe-core formation timeline



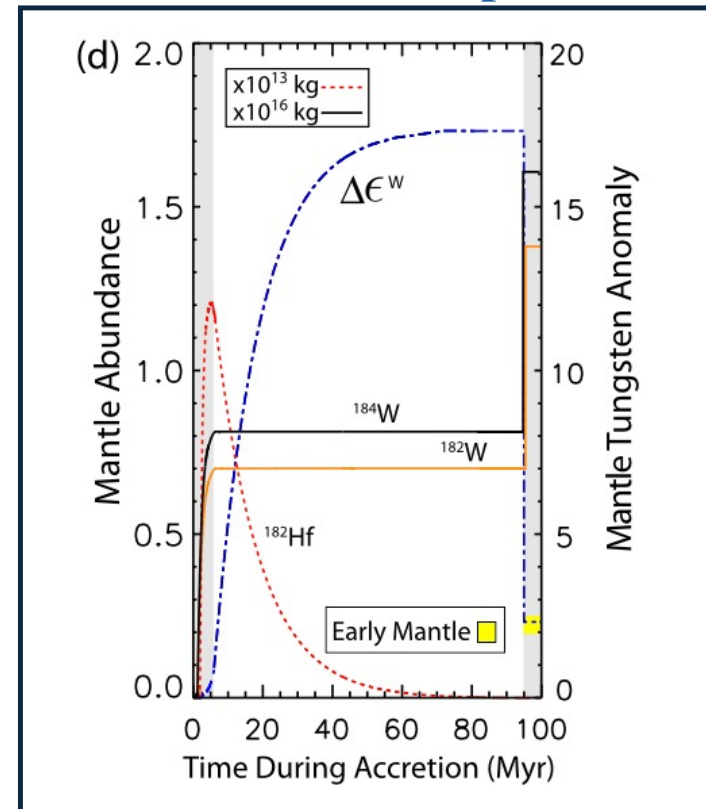
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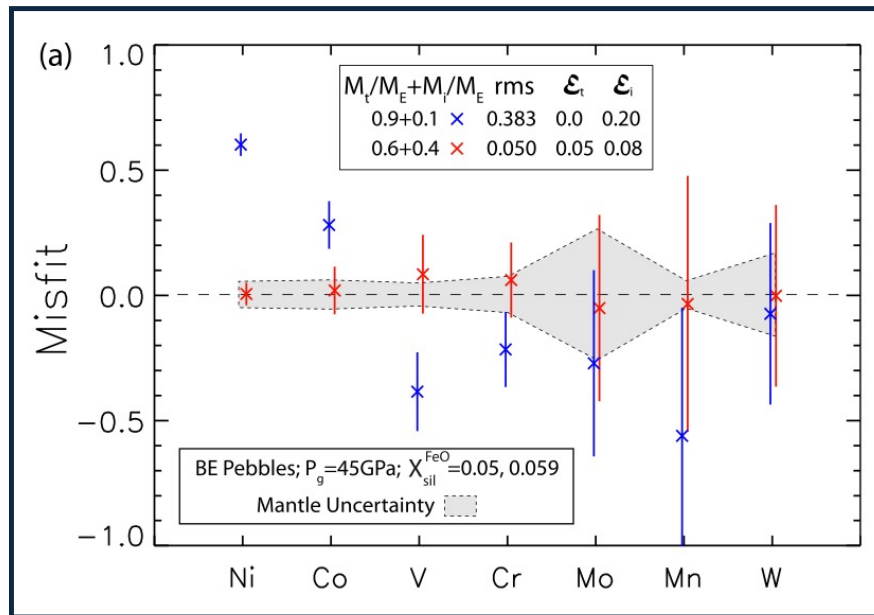
Olson and Sharp. (2023); Johansen et al. (2023)

Pebble + impact



Earth's siderophile composition with pebbles

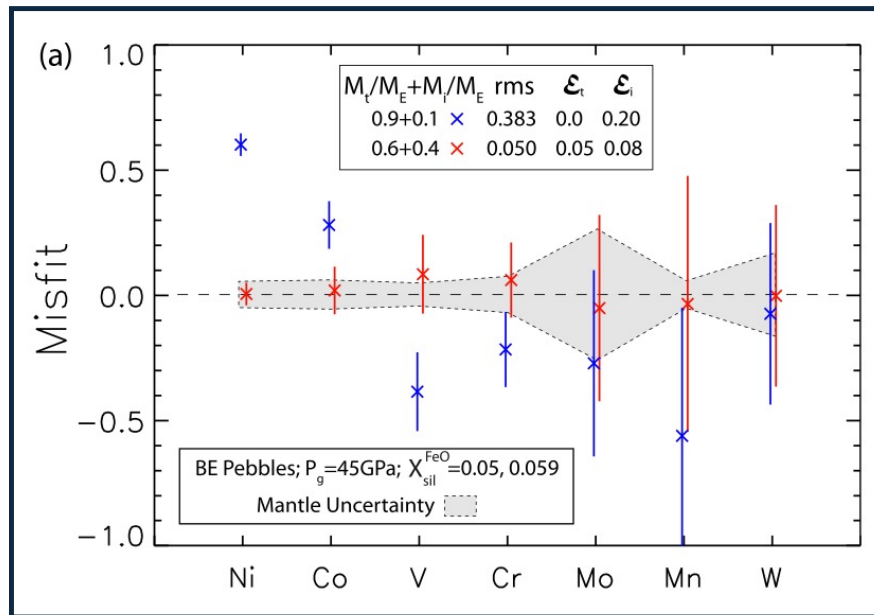
A Pebble + impact



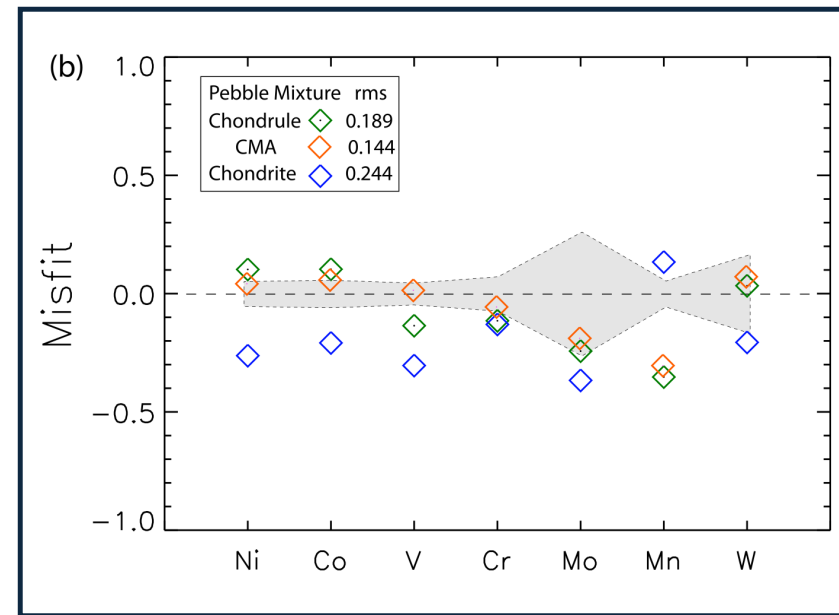
Olson et al. (2025)

Earth's siderophile composition with pebbles

A Pebble + impact

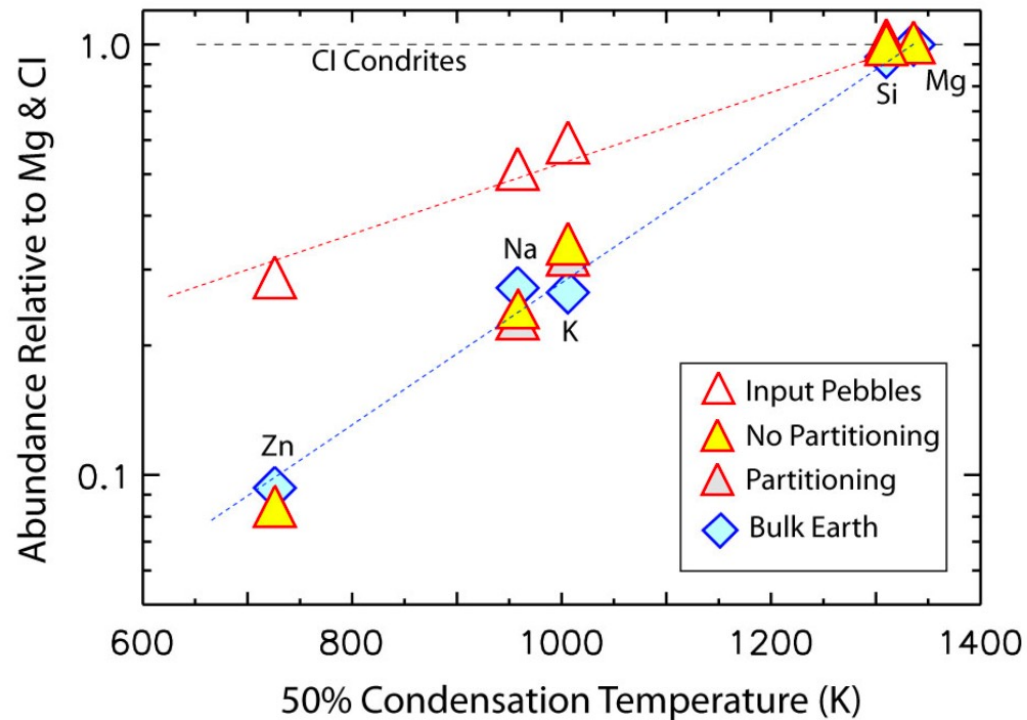


B Chondrite vs pebbles



Olson et al. (2025)

Earth's moderately volatile depletion



Depletion is best explained by a combination of (0.65+0.35) Earth mass (proto-Earth+Theia).

Olson et al. (2026)

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4. Match Earth's moderately volatile composition (Si (as SiO), Na, K, and Zn)? ☒

Summary

1. Mixtures of **chondritic components explain Earth's major element, siderophile element, and moderately volatile compositions**; mixtures of bulk chondrites do not.
2. The fits of the above variety of elements imply **a hybrid accretion scenario for Earth**. A few million years of pebble accretion leading to the formation of a $\sim 0.6\text{--}0.7$ Earth-mass proto-Earth and a nearby $\sim 0.4\text{--}0.3$ Earth-mass impactor (Theia), followed by late collisions completing the assembly.
3. In the pebble accretion scenario,
 - Vaporized pebbles produce 100s of ocean masses of water in the atmosphere. **Water on Earth may not require any exogenous source** (water-rich comets, asteroids, etc.).
 - **The nebular exhaust explains volatile depletion.**